

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0301

Alvin 138 kV
May 21, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0301. The Interconnection Customer proposes a 167.0 MW solar photovoltaic facility interconnecting at the Alvin 138 kV substation in the Virginia area. The request was received on January 08, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 189 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,476,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Chatom / Purdy 138kV is overloaded to 125.6% of its 279 MVA emergency rating for the loss of the Frito Lay / Dpc Prairie Du Chien 138kV Line. The project contributes 28.5% DFAX (47.6 MW) to this overload.

The St Louis / Nichols 138kV is overloaded to 119.4% of its 185 MVA emergency rating for the loss of the Purdy / Page 138kV. The project contributes 23.1% DFAX (38.5 MW) to this overload.

The Dpc Prairie Du Chien / Frito Lay 138kV Line is overloaded to 116.2% of its 233 MVA emergency rating for the loss of the Chatom / Purdy 138kV. The project contributes 20.9% DFAX (34.9 MW) to this overload.

For the loss of Harrisonburg / Sans Souci Substation 138kV, the Alvin / Nuclear Fuels 138kV loads to 112.1% of its 201 MVA rating. Project DFAX: 23.1%.

S Belleville / Nichols Substation 138kV: 109.8% loading (197 MVA) following the outage of the Purdy / Page 138kV; DFAX = 23.1%.

For the loss of American / Urban 345kV, the Tes Filer Cit Station Substation / Bridge River 138kV loads to 106.4% of its 206 MVA rating. Project DFAX: 7.8%.

Innovative Solar 31 / Kapstone 138kV: 104.4% loading (249 MVA) following the outage of the Grimes / Louisa Generation Facility 138kV; DFAX = 14.8%.

Page / Purdy 138kV L: 103.2% loading (261 MVA) following the outage of the Nichols / St Louis 138kV; DFAX = 21.3%.

Innovative Solar 31 / Pratt'S 138kV: 103.0% loading (247 MVA) following the outage of the Louisa Generation Facility / Grimes 138kV line; DFAX = 14.8%.

Page / Cedarville Jct 138kV Line: 102.0% loading (252 MVA) following the outage of the St Louis / Nichols 138kV Line; DFAX = 21.3%.

The Cedarville Jct / Detour 138kV line is overloaded to 101.8% of its 248 MVA emergency rating for the loss of the Nichols / St Louis 138kV. The project contributes 21.3% DFAX (35.6 MW) to this overload.

For the loss of Fluvanna P.S Substation / Louisa Generation Facility 345kV, the Louisa Generation Facility 345-138 kV Transformer Bank loads to 100.9% of its 601 MVA rating. Project DFAX: 12.1%.

Northpoint / S Belleville Substation 138kV L: 100.7% loading (208 MVA) following the outage of the Purdy / Page 138kV; DFAX = 23.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chatom / Purdy 138kV	Rebuild line with higher-capacity conductor	\$7,526,000	36% (\$2,728,000)
St Louis / Nichols 138kV	Rebuild line with higher-capacity conductor	\$9,273,000	26% (\$2,384,000)
Dpc Prairie Du Chien / Frito Lay 138kV Line	Rebuild line with higher-capacity conductor	\$3,143,000	32% (\$1,016,000)
Alvin / Nuclear Fuels 138kV Line	Rebuild line with higher-capacity conductor	\$7,348,000	47% (\$3,436,000)
S Belleville / Nichols Substation 138kV	Reconductor line	\$1,351,000	26% (\$347,000)
Tes Filer Cit Station Substation / Bridge River 138kV	Reconductor line	\$1,357,000	19% (\$257,000)
Innovative Solar 31 / Kapstone 138kV	Reconductor line	\$2,386,000	83% (\$1,985,000)
Page / Purdy 138kV	Reconductor line	\$998,000	36% (\$362,000)
Innovative Solar 31 / Pratts 138kV	Reconductor line	\$1,636,000	83% (\$1,360,000)
Page / Cedarville Jct 138kV Line	Reconductor line	\$3,690,000	36% (\$1,339,000)
Detour / Cedarville Junction Substation 138kV	Reconductor line	\$3,998,000	36% (\$1,451,000)
Louisa Generation Facility 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,846,000	6% (\$986,000)
Northpoint / S Belleville Substation 138kV L	Reconductor line	\$3,208,000	26% (\$825,000)
	Total allocated to project		\$18,476,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.